EX. NO: 4 — VIGENERE CIPHER

Aim:

To implement the Vigenere Cipher substitution technique using C.

Algorithm:

1. Read plaintext and key.
2. Repeat the key to match plaintext length.
3. Add plaintext and key values.
4. Apply modulo 26.
5. Display encrypted text.
6. Reverse the operation for decryption.

Program:

```
#include <stdio.h>
#include <string.h>
#include <ctype.h>
int main()
{
    char text[100], key[50];
    int i, j, len;
    printf("Enter plain text: ");
    scanf("%s", text);
    printf("Enter key: ");
    scanf("%s", key);
    len = strlen(key);
    printf("Encrypted text: ");
    for(i = 0, j = 0; text[i] != '\0'; i++, j++)
    {
        if(j == len)
            j = 0;
```

```
        printf("%c",
            ((toupper(text[i])-'A') +
             (toupper(key[j])-'A')) % 26 + 'A');
    }
    return 0;
}
```

Output:

Result:

Thus, the Vigenere Cipher substitution technique was implemented successfully using C.